NCERT Summary: Topic 03: Consumer Behaviour & Demand

Source Integration

- NCERT Class 12 Introductory Microeconomics, Chapter 2: *Theory of Consumer Behaviour*

Core Factual & Conceptual Architecture

1. The Consumer's Optimization Problem

- Objective: Maximize satisfaction subject to an exogenous budget constraint.
- Two goods abstraction: Quantities of Good 1 (x_1) and Good 2 (x_2) forming consumption bundle (x_1, x_2).
- Dual analytical frameworks:
  - **Cardinal Utility Theory**: Measurable in utils; additive utility function.
  - **Ordinal Utility Theory**: Ranking of commodity bundles based on preference ordering (Hicks-Allen framework).

2. Cardinal Utility Foundations

- **Total Utility (TU)**: Cumulative subjective satisfaction derived from consuming a given aggregate quantity: $TU_n = \sum MU_i$.
- **Marginal Utility (MU)**: Incremental satisfaction from one additional unit: $MU = \frac{\Delta TU}{\Delta Q} = TU_n - TU_{n-1}$.
- **Law of Diminishing Marginal Utility (Gossen's First Law)**: As consumption of a homogeneous good increases consecutively without time lags, MU systematically declines.
- **Consumer Equilibrium (Single Good)**: Attained where $MU_x = P_x$ (in monetary terms), or $\frac{MU_x}{P_x} = MU_m$ (marginal utility of money).

3. Ordinal Utility & Indifference Curve (IC) Mechanics

- **Axiom of Monotonic Preferences**: A rational consumer strictly prefers a bundle with more of at least one good and no less of another (non-satiation).
- **Indifference Curve (IC)**: Locus of consumption bundles yielding identical aggregate subjective utility.
- **Properties of Indifference Curves**:
  1. Downward sloping from left to right (negative slope reflecting trade-off).
  2. Convex to the origin due to the **Law of Diminishing Marginal Rate of Substitution (MRS)**.
  3. Higher indifference curves represent strictly higher utility levels.
  4. Two indifference curves can never intersect (violation of transitivity and consistency).
- **Marginal Rate of Substitution (MRS_{xy})**: Slope of the IC. The rate at which the consumer is willing to substitute Good 2 for Good 1: $MRS = -\frac{\Delta x_2}{\Delta x_1} = \frac{MU_1}{MU_2}$.

4. The Budget Constraint

- **Budget Set**: All affordable consumption bundles: $P_1 x_1 + P_2 x_2 \le M$ (where M is nominal income).
- **Budget Line Equation**: $P_1 x_1 + P_2 x_2 = M \implies x_2 = \frac{M}{P_2} - \frac{P_1}{P_2} x_1$.
- **Slope of the Budget Line**: $-\frac{P_1}{P_2}$ (market-determined trade-off ratio / opportunity cost).
- **Shifts vs. Rotations**:
  - Income change (ΔM) with constant prices: Parallel outward/inward shift.
  - Price change (ΔP_1 or ΔP_2): Pivot/rotation along the affected axis.

5. Consumer Equilibrium (Ordinal Framework)

- Equilibrium condition: Budget line is strictly tangent to the highest attainable Indifference Curve.
- Tangency rule: Slope of IC = Slope of Budget Line: $$MRS_{12} = \frac{P_1}{P_2} \implies \frac{MU_1}{MU_2} = \frac{P_1}{P_2} \implies \frac{MU_1}{P_1} = \frac{MU_2}{P_2}$$
- The consumer's subjective valuation (willingness to trade) perfectly equates to the market's objective valuation (price ratio).

6. Demand Functions & Cross-Relationships

- **Law of Demand**: Inverse relationship between own-price and quantity demanded, *ceteris paribus*.
- **Income Classification**:
  - *Normal Goods*: Positive income effect ($\frac{\Delta Q}{\Delta M} > 0$).
  - *Inferior Goods*: Negative income effect ($\frac{\Delta Q}{\Delta M} < 0$).
- **Cross-Price Relationships**:
  - *Substitutes*: Positive cross-price elasticity ($\frac{\Delta Q_A}{\Delta P_B} > 0$, e.g., Tea and Coffee).
  - *Complements*: Negative cross-price elasticity ($\frac{\Delta Q_A}{\Delta P_B} < 0$, e.g., Car and Petrol, Tea and Sugar).
- **Movement along vs. Shift**:
  - *Change in Quantity Demanded*: Movement along a stationary curve caused solely by own-price variation.
  - *Change in Demand*: Structural shift of the entire curve driven by income, preferences, or prices of related commodities.

7. Price Elasticity of Demand (E_d)

- Point Elasticity: $E_d = \frac{\% \Delta Q}{\% \Delta P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$.
- **Linear Demand Curve Elasticity**:
  - Midpoint: $|E_d| = 1$
  - Upper half (towards price intercept): $|E_d| > 1$ (approaching ∞ at vertical axis)
  - Lower half (towards quantity intercept): $|E_d| < 1$ (approaching 0 at horizontal axis)
- **Total Outlay / Expenditure Method**:
  - $|E_d| > 1$: Price and Total Expenditure move in opposite directions.
  - $|E_d| < 1$: Price and Total Expenditure move in the same direction.
  - $|E_d| = 1$: Total Expenditure remains invariant to price movements.

✍ Notes & Updates